Model	GSM8K	MATH-500	AIME2024	AIME2025	AMC2023	OlympiadBench	AVG
LLaMA-3.1-8B-Base [9]	7.58	3.20	0.00	0.00	4.22	1.19	2.70
+ 20K R1-Distill-Data	63.38	30.60	0.21	0.42	14.22	8.30	19.52
+ 20K OSS-Distill-Data	75.89	54.20	4.38	6.46	37.34	23.85	33.69
+ 20K QwQ-Distill-Data	64.53	32.20	2.92	0.42	16.72	8.89	20.95
Llama-3.1-8B-Instruct [9]	75.89	35.20	4.17	1.04	23.59	12.00	25.32
+ 20K R1-Distill-Data	79.91	60.60	2.50	3.88	33.13	23.85	33.98
+ 20K OSS-Distill-Data	79.00	60.80	10.83	7.71	47.03	30.22	39.27
+ 20K QwQ-Distill-Data	82.41	60.80	4.38	8.33	32.97	25.48	35.73

Table 1 Results across six benchmarks. Full results are reported in Table 5 in the Appendix.

Self-Exploration gently links different long-distance clusters by Van der Waals Forces. In contrast to local stabilization (Fig. 7 (c)), exploration transitions act as loose links between otherwise separated clusters. They show much larger step-to-step distances, with an average trajectory length of 5.32 in the 3D t-SNE projection. Together, these results suggest that effective Long CoT reasoning is not a simple linear chain; instead, it forms a folded, domain-structured topology consistent with the "logical folding" hypothesis.

4.4 Attention ⇔ Energy Level of Bonds

In physical chemistry, the behavior probability with energy level E_i at temperature T follows the Boltzmann distribution:

$$P(\text{state}_i) = \frac{\exp\big(-E_i/k_B T\big)}{\sum_j \exp\big(-E_j/k_B T\big)}. \tag{1}$$

In Transformers, the attention weight α_{ij} of the i-th token to the j-th token is:

$$\alpha_{ij} = \frac{\exp\big(q_i \cdot k_j/\sqrt{d_k}\big)}{\sum_l \exp\big(q_i \cdot k_l/\sqrt{d_k}\big)}. \tag{2}$$

The correspondence follows by defining attention energy $(E) \leftrightarrow (-q \cdot k)$, implies lower E_{ij} and thus higher behavior probability.

Formally, our analysis only relies on the observation that attention weights form a Gibbs–Boltzmann distribution over negative logits. We therefore use the term "energy" to denote reparameterized logits and study how their expectations differ across behavior types.

We then compare attention energy across Long CoT transition types. Fig. 8 shows distinct distributions for deep reasoning, reflection, and exploration. Deep reasoning exhibits the largest effective chemical-bond energy D_d. Reflection is intermediate, whereas exploration shows the weakest effective bond energy. This ordering and the relative proportions are consistent across models, supporting the hypothesis that a bond-like mechanism broadly links these reasoning behaviors.

Takeaway 2

- Long CoT reasoning exhibits stable structures across models, with reasoning topologies converging.
- Semantic isomers, reasoning chains with identical concepts but different logical bonds, succeed or fail based on bond structure, not surface keywords.
- Three distinct logical bonds drive CoT structure: reflection folds back to prior clusters, deep reasoning creates stable local domains, and exploration bridges distant concepts, each with characteristic attention energy profiles matching Boltzmann-like distributions.
- SFT learns reasoning structure rather than surface keywords, determining Long CoT capability.

5 Feature: Effective Semantic Isomers

We now shift to examine the failure of surficial distillation: similar conceptual atoms can be linked by different bonds, producing completely different reasoning chains. We call these as **Semantic Isomers**. Mathematically,

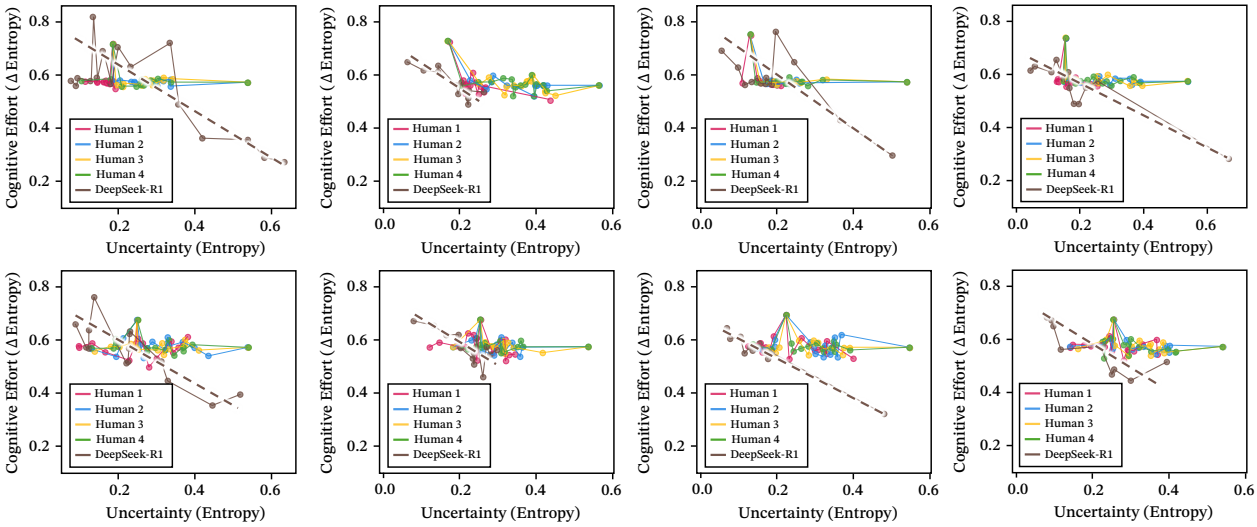


Figure 10 Information flow analysis for humans and reasoning models.

a semantic isomer of $\mathcal{D}$ is any $\mathcal{D}'$ whose $(P_{\mathcal{D}'}, \pi_{\mathcal{D}'})$ lies near $(P_{\mathcal{D}}, \pi_{\mathcal{D}})$ under a suitable divergence $D(\cdot \| \cdot)$. Then, we will analyze how they form, are learned, and sometimes destabilize in Long CoT.

5.1 Bond structure of Semantic Isomers is the key to Long-CoT learning

Well-structured Semantic Isomers can be effective for Long-CoT learning. To assess whether well-structured Semantic Isomers can enhance Long-CoT learning, we constructed a Long CoT dataset by distilling from advanced reasoning LLMs. As shown in Table 1, models exhibit consistent performance gains when trained on these variants (distribution correlation ∼0.9). This suggests that, within a certain range, multiple effective "allotropic" variants of reasoning keys can exist.

The model has multiple effective Semantic Isomers, and slight differences can significantly affect the results. As shown in Table 1, the structural correlation between R1- and OSS-derived chains can reach 0.95, yet for some models performance with R1-based chains drops by more than 10%. This gap indicates the presence of multiple near-optimal Semantic Isomers that are nonetheless fragile, where small changes in the distribution can cause large performance losses.

Simulating effective Semantic Isomer structures is the key to ICL distillation. We analyze three ICL setups to simulate Long CoT chains by Qwen2.5-32B-Instruct: demonstrations chosen at random; demonstrations with closely aligned reasoning-key distributions (correlation ∼0.9); and demonstrations with clearly mismatched distributions (correlation < 0.8). As shown in Fig. 9, substantial performance gains emerge only when demonstrations are constructed to match a specific target distribution of reasoning keys, thereby yielding an appropriate "allotrope" of that distribution.

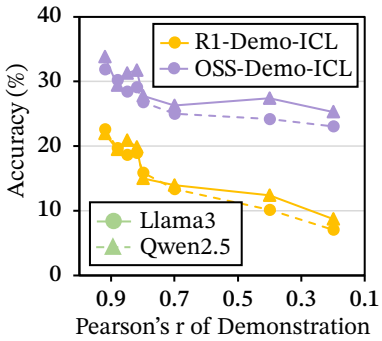
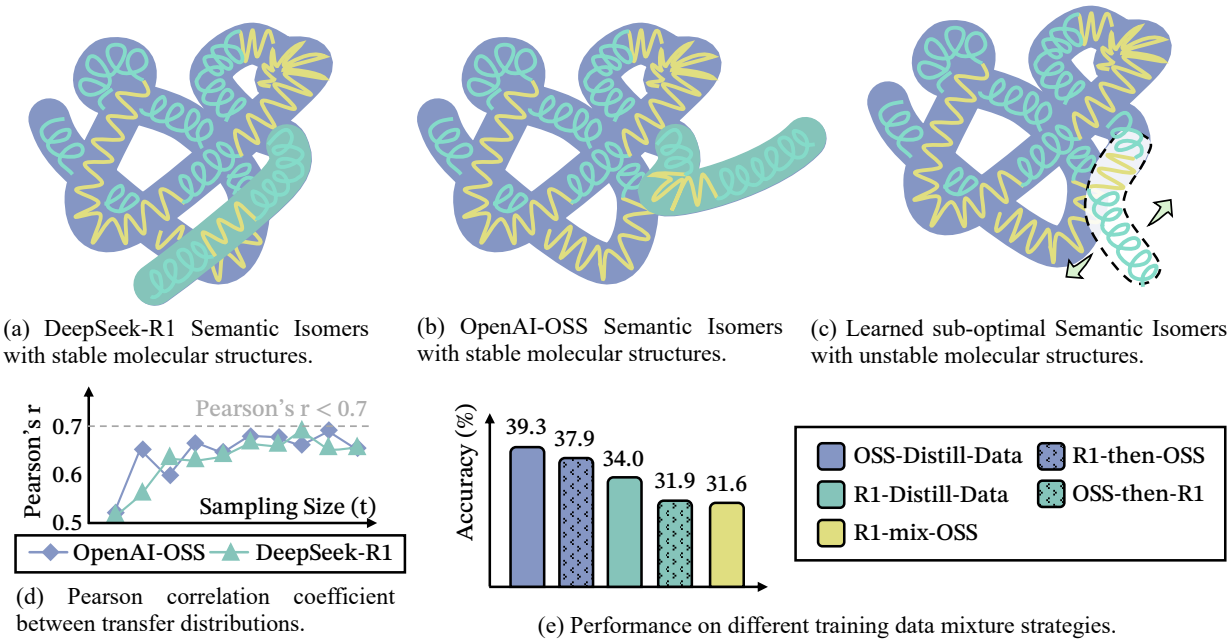


Figure 9 The performance of LLama-3.1-8B-Instruction under three different distillation setups from ICL-augmented Qwen2.5-32B.

5.1.1 Not all bonds in isomers are effective.

To clarify the nature of semantic isomers, we examine which bond structures yield effective reasoning configurations. We hypothesize that functional viability depends on specific bond distributions: despite sharing identical conceptual nodes, incompatible configurations disrupt information exchange. For instance, excessive exploration bonds cause fragmented reasoning, whereas overemphasized deep reasoning bonds create rigid chains unable to adapt to new inputs.Details are provided in Appendix D.3.



(a) DeepSeek-R1 Semantic Isomers with stable molecular structures.

(b) OpenAI-OSS Semantic Isomers with stable molecular structures.

(c) Learned sub-optimal Semantic Isomers with unstable molecular structures.

(d) Pearson correlation coefficient between transfer distributions.

(e) Performance on different training data mixture strategies.

Figure 11 Conflict Learning between Two Stable Molecular Structures.

Effective reasoning bond distribution influences the information divergence speed in reasoning dynamics. To assess this, we compared the reasoning dynamics of R1 models with human cognition in an information phase space [10]. Mechanistically, LLMs update by maximizing rewards and reducing entropy, whereas human reasoning is additionally constrained by semantic coherence and social feedback. Consequently, machine reasoning converges through accumulated gradient updates, whereas human reasoning stabilizes through iterative self-monitoring and social calibration. As shown in Figure 10, we tracked reasoning unfolding over extended chains in logical deduction tasks. Humans typically exhibit nearly uniform forward information gains (81.3% of cases show changes < 0.1), corresponding to a near-zero slope in phase space. In contrast, R1 models display accelerating informativeness (76.1% of cases show absolute changes > 0.1), progressing from low entropy to rapid convergence. These patterns indicate a fundamental difference in how R1 models and humans integrate information over time.

Effective reasoning bonds cause metacognitive oscillation and alignment. We identify the core rationale for this difference as a "metacognitive oscillation" in LLMs. Their responses alternate between high-entropy divergent exploration (slope > 0.6, Δentropy > 0.05) and stable convergent validation, a dynamic less pronounced in the uniform entropy profiles of humans. Case studies (Figure 10) confirm that R1 models utilize self-reflective revision to adjust reasoning paths against uncertainty. We hypothesize that aligning training objectives with these behavioral structure distributions can narrow the gap between machine and human reasoning dynamics.

5.2 Conflict between Two Stable Structures

Understanding how distinct reasoning structures interact reveals fundamental limits of complex cognitive systems. As shown in Fig. 11 (a–c), forcibly fusing stable molecular isomers disrupts their backbone; analogously, combining incompatible reasoning frameworks breaks global logical coherence.

Learning two heterogeneous stable structures at the same time will lead to structural chaos in the model. As shown in Fig. 11 (d), we test this by jointly activating two highly correlated ($r \approx 0.9$) reasoning chains from DeepSeek-R1 and OpenAI-OSS. Despite their similarity, co-activation prevents the model from converging to a single stable behavioral mode: it produces molecular bond distributions that fluctuate across samples and deviate from those characteristic of either OSS or R1. Consistent with this instability, the self-correlation of the jointly activated model does not exceed 0.8.